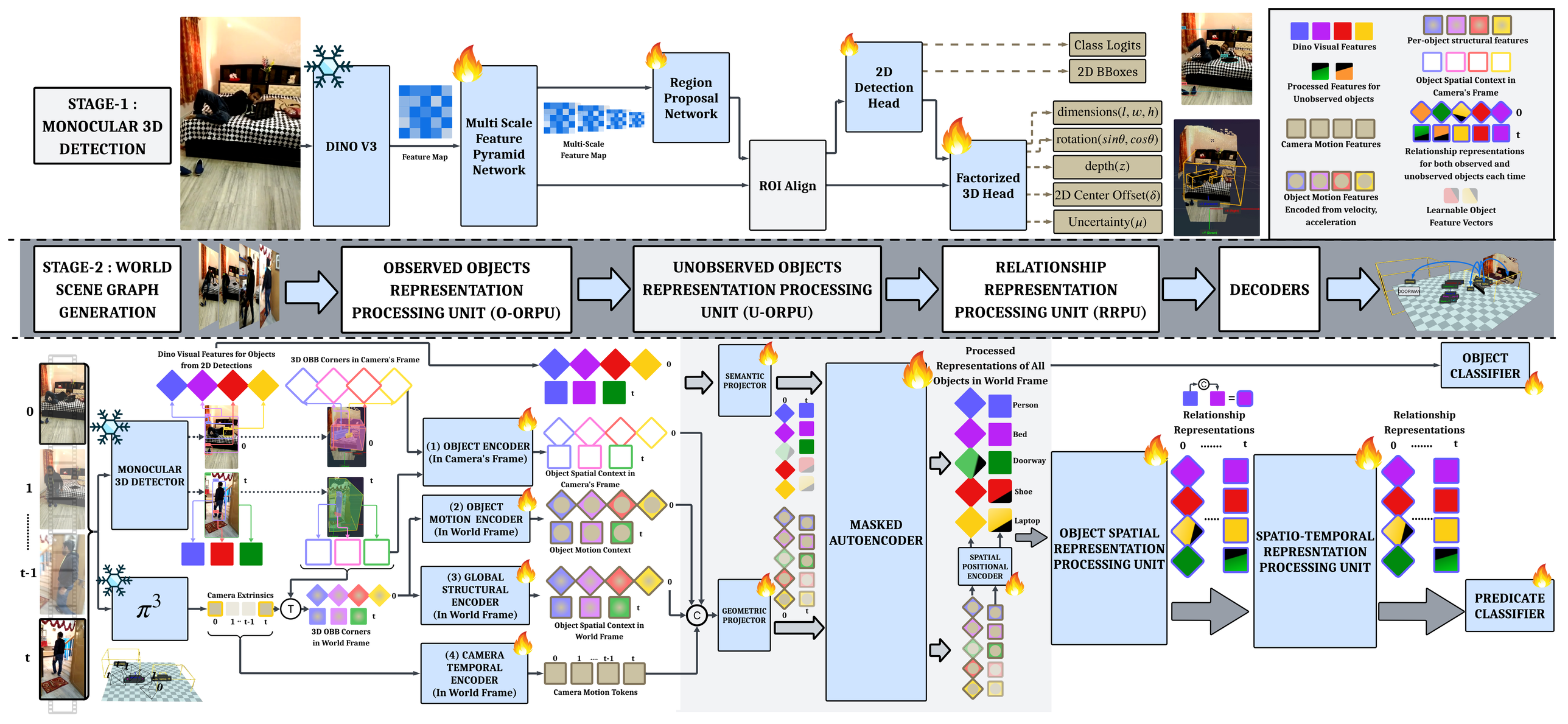
STAGE-1 : MONOCULAR 3D DETECTION
DINO V3
Feature Map
Multi Scale Feature Pyramid Network
Multi-Scale Feature Map
Region Proposal Network
ROI Align
2D Detection Head
Class Logits
2D BBoxes
Factorized 3D Head
dimensions(l, w, h)
rotation($sin\theta, cos\theta$)
depth(z)
2D Center Offset(δ)
Uncertainty(μ)
Dino Visual Features
Per-object structural features
Processed Features for Unobserved objects
Object Spatial Context in Camera's Frame
Camera Motion Features
Relationship representations for both observed and unobserved objects each time
Object Motion Features Encoded from velocity, acceleration
Learnable Object Feature Vectors
STAGE-2 : WORLD SCENE GRAPH GENERATION
OBSERVED OBJECTS REPRESENTATION PROCESSING UNIT (O-ORPU)
UNOBSERVED OBJECTS REPRESENTATION PROCESSING UNIT (U-ORPU)
RELATIONSHIP REPRESENTATION PROCESSING UNIT (RRPU)
DECODERS
DOORWAY
Dino Visual Features for Objects from 2D Detections
3D OBB Corners in Camera's Frame
MONOCULAR 3D DETECTOR
π^3
Camera Extrinsics
3D OBB Corners in World Frame
(1) OBJECT ENCODER (In Camera's Frame)
Object Spatial Context in Camera's Frame
(2) OBJECT MOTION ENCODER (In World Frame)
Object Motion Context
(3) GLOBAL STRUCTURAL ENCODER (In World Frame)
Object Spatial Context in World Frame
(4) CAMERA TEMPORAL ENCODER (In World Frame)
Camera Motion Tokens
SEMANTIC PROJECTOR
GEOMETRIC PROJECTOR
MASKED AUTOENCODER
Processed Representations of All Objects in World Frame
Person
Bed
Doorway
Shoe
Laptop
SPATIAL POSITIONAL ENCODER
OBJECT CLASSIFIER
OBJECT SPATIAL REPRESENTATION PROCESSING UNIT
Relationship Representations
SPATIO-TEMPORAL REPRESNTATION PROCESSING UNIT
PREDICATE CLASSIFIER